Smart Library Management & Tracking System

System Architecture, Hardware Tracking & Functional Project Specification

1. Executive Summary

The **Smart Library Management & Tracking System (SLMTS)** is an IoT-connected, automated platform designed to overhaul traditional manual library operations. By integrating **High-Frequency RFID (ISO 15693 / 28560)**, anti-theft **Electronic Article Surveillance (EAS)** pedestals, self-checkout touch kiosks, 24/7 smart book return chutes, and real-time seat reservation with presence sensing, the system eliminates checkout queues, prevents book theft, and provides full transparency for students, faculty, and administrative staff.

2. Core System Modules

2.1 Book Inventory & Asset Tracking

- **Dual Identification (RFID + Barcode):** HF RFID tags (13.56 MHz, ISO 15693 / 28560) allow non-line-of-sight scanning of multiple books simultaneously on reader pads.
- **Anti-Theft EAS Security Gates:** RFID tags carry an EAS/AFI security byte. When checked out legitimately, the kiosk deactivates this byte. Unauthorized passage triggers physical alarm gates and camera snapshots.
- **ISBN Auto-Enrichment:** Integrated with Google Books and OpenLibrary APIs to instantly pull title, author, cover image, and Dewey Decimal Classification upon barcode scan.
- **Handheld Audit Wand:** Enables library staff to scan whole shelf racks at 50+ books/sec to pinpoint misplaced or missing items.

2.2 Student, Faculty & Staff Management

- **Role-Based Access Control (RBAC):** Discrete permission boundaries for Super Admin, Librarians, Faculty, and Students.
- **Virtual Digital ID Card:** Mobile app generates dynamic, anti-screenshot QR codes and NFC card emulation for contactless gate entry and kiosk authentication.
- **Differentiated Borrowing Rules:** Configurable borrowing quotas (e.g. Undergrads: 4 books/14 days; Faculty: 15 books/90 days).

2.3 Automated Circulation & Self-Service

- **Self-Checkout Kiosk:** Patron authenticates, stacks up to 5 books on the multi-tag RFID pad, system validates loans in one atomic transaction, deactivates EAS bits, and emails receipts.
- **24/7 Smart Book Chute:** Exterior return drop box equipped with internal RFID reader automatically checks in returned books, reactivates EAS protection, and notifies the patron.
- **Automated Fine Engine:** Calculates late penalties on a per-day schedule, automatically exempting institutional holidays, weekends, and exam breaks.

2.4 Smart Seat & Study Room Booking

- **Interactive 2D Live Floor Map:** Real-time visualization of vacant, booked, and occupied carrels and discussion suites.
- **15-Minute No-Show Auto-Release:** Booked seats must be claimed via desk QR scan within 15 minutes, or the slot is auto-cancelled and freed.

- **IoT Smart Room Locks:** Electronic locks grant entry only during active, confirmed reservation windows.

3. Recommended Technology Stack

Layer	Technology Selection	Purpose / Advantage
Web Portal	Next.js 14+ / React / Tailwind CSS	SEO-optimized public catalog and reactive staff portal
Mobile App	React Native (Expo) / Flutter	Virtual student ID card, camera QR scanning, push alerts
Kiosk Interface	Electron / Flutter Desktop	Direct low-level USB/Serial access to RFID pads & printers
Backend Core	Node.js (NestJS) / Python (FastAPI)	Modular service architecture, WebSockets, REST APIs
Primary DB	PostgreSQL 16	ACID guarantees for circulation loans and JSONB metadata
Cache & Locks	Redis	Distributed locking for seat reservations and fast sessions
Search Engine	Meilisearch	Sub-20ms typo-tolerant fuzzy catalog search
Hardware / IoT	ESP32 / Raspberry Pi / HF RFID	Smart chute sensors, EAS gate alarms, smart door relays

4. Implementation Roadmap

Phase	Key Deliverables	Focus Area
Phase 1	Database schema, RBAC Auth, Book Catalog CRUD with ISBN auto-fill	Core Foundation
Phase 2	Circulation desk (Issue/Return), Fine calculator, Patron loan matrix	Staff Workflows
Phase 3	Self-checkout kiosk, Smart drop chute, EAS gate alarm bridge (RFID)	Hardware & IoT
Phase 4	Interactive 2D seat map, QR check-in auto-release, Mobile app	Space & Patron App

5. Security, Privacy & Reliability

- **Patron Privacy (FERPA / GDPR):** Historical borrowing logs are anonymized once items are returned and settled to safeguard user reading confidentiality.
- **Hardware Independence:** EAS security pedestals operate autonomously at the hardware level, sounding alarms even during network drops.
- **Offline Kiosk Operation:** Kiosks queue checkouts locally and synchronize with the central database when connection restores.